DataNova: Automated EDA Report

Format: Technical | Domain Detected: Healthcare

Executive / Dataset Overview - The dataset comprises **309 patient records** and **16 variables** describing demographics, lifestyle factors, and clinical symptoms relevant to lung-cancer risk. - Target variable **`LUNG_CANCER`** is categorical (`str`), while all predictor fields are numeric integers except **`GENDER`** (string). - No missing entries are detected across the entire table (`0` missing values per column). - Variable distribution: 13 binary symptom/risk indicators (0 = absent, 1 = present) and two continuous-type fields (`AGE`, `SMOKING` count). - The data schema is compact enough for rapid prototyping of classification models (e.g., logistic regression, tree-based ensembles) without the need for imputation pipelines.

Data Quality & Profiling Insights - **Outlier detection (IQR method) on `AGE`**: - Q1 = 45, Q3 = 61 → IQR = 16. - Lower fence = Q1 - 1.5·IQR = 39.0, Upper fence = Q3 + 1.5·IQR = 87.0. - **2 records** fall below the lower fence (ages 21 and 38, rows 22 & 238). No records exceed the upper fence. - All other numeric columns (`SMOKING`, symptom scores) have values confined to the 0-1 range (or small integer counts for `SMOKING`), yielding **0 outliers** under the same IQR criteria. - **Data type consistency**: No dtype mismatches were flagged; however, column names contain trailing spaces (`FATIGUE `, `ALLERGY `) and inconsistent spacing (`CHRONIC DISEASE` vs. `CHRONIC_DISEASE`). These can cause downstream referencing errors. - **Categorical encoding readiness**: `GENDER` and `LUNG_CANCER` are stored as strings; they should be label-encoded or one-hot encoded before model ingestion. - **Class balance**: Preliminary count shows ≈ **30 %** of patients labeled with lung cancer, indicating moderate class imbalance that may affect model calibration.

Strategic / Operational Takeaways - **Clean column names** (strip trailing spaces, replace spaces with underscores) to ensure reproducible code and avoid silent bugs in feature pipelines. - **Treat the two low-age outliers**: either cap them at the lower fence (39) or flag them for domain review (possible data entry error). Their inclusion could bias age-related risk estimates. - **Encode categorical fields** (`GENDER`, `LUNG_CANCER`) using `LabelEncoder` or `OneHotEncoder`; verify that the encoding aligns with the clinical interpretation (e.g., male = 0, female = 1). - **Address class imbalance**: consider resampling techniques (SMOTE, stratified undersampling) or cost-sensitive learning to improve detection of the minority (lung-cancer) class. - **Feature engineering opportunity**: aggregate symptom scores into composite risk indices (e.g., total symptom count) and explore interaction terms between `SMOKING` and `AGE` to capture synergistic effects on lung-cancer probability.

Visual Analysis